

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
7th Semester of B. Pharm. Examination
University Theory Examination April/May 2016
PH404 Pharmaceutical Analysis III

Date: 12.04.2016, Tuesday

Time: 1:30 p.m to 4:30 p.m

Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

Q 1 Attempt any **TWO** of the following

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|---|--|----|
| A | 1) What is the fundamental law governing UV absorption and why do molecules deviate from the law? | 05 |
| | 2) Explain any one detector used in UV with suitable diagram. | 05 |
| B | 1) Explain the principle of IR spectroscopy | 05 |
| | 2) Explain the different sampling techniques employed in Infrared Spectrophotometry. | 05 |
| C | 1) Explain the types of fluorescence quenching with examples and give the application of fluorimetry in pharmaceuticals. | 07 |
| | 2) Draw a neat labeled Jablonski Diagram. | 03 |

Q 2 Attempt any **TWO** of the following

- | | | |
|---|---|----|
| A | What is FT-IR? Explain the principle and Instrumentation of FTIR with a neat diagram. | 10 |
| B | 1) Explain double beam UV spectrophotometer with a neat, labeled, block diagram. | 06 |
| | 2) Explain the application of UV in single component analysis. | 04 |
| C | a) Write different types of transitions in UV. | 03 |
| | b) What are the factors affecting $\lambda_{\max}$ of a compound? | 07 |

SECTION- II

Q 3 Attempt any **TWO** of the following

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|---|---|----|
| A | (1) What is HPTLC? Differentiate HPTLC and TLC. | 05 |
| | (2) Enlist the steps involved in HPTLC. | 03 |
| | (3) Explain the term chamber saturation and its importance. | 02 |